

ACE - LENSING EMULATOR

Accurate cosmological emulator for the probability distribution function of gravitational lensing of point sources.

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PhD Candidate · UFES, Brazil

July 2026



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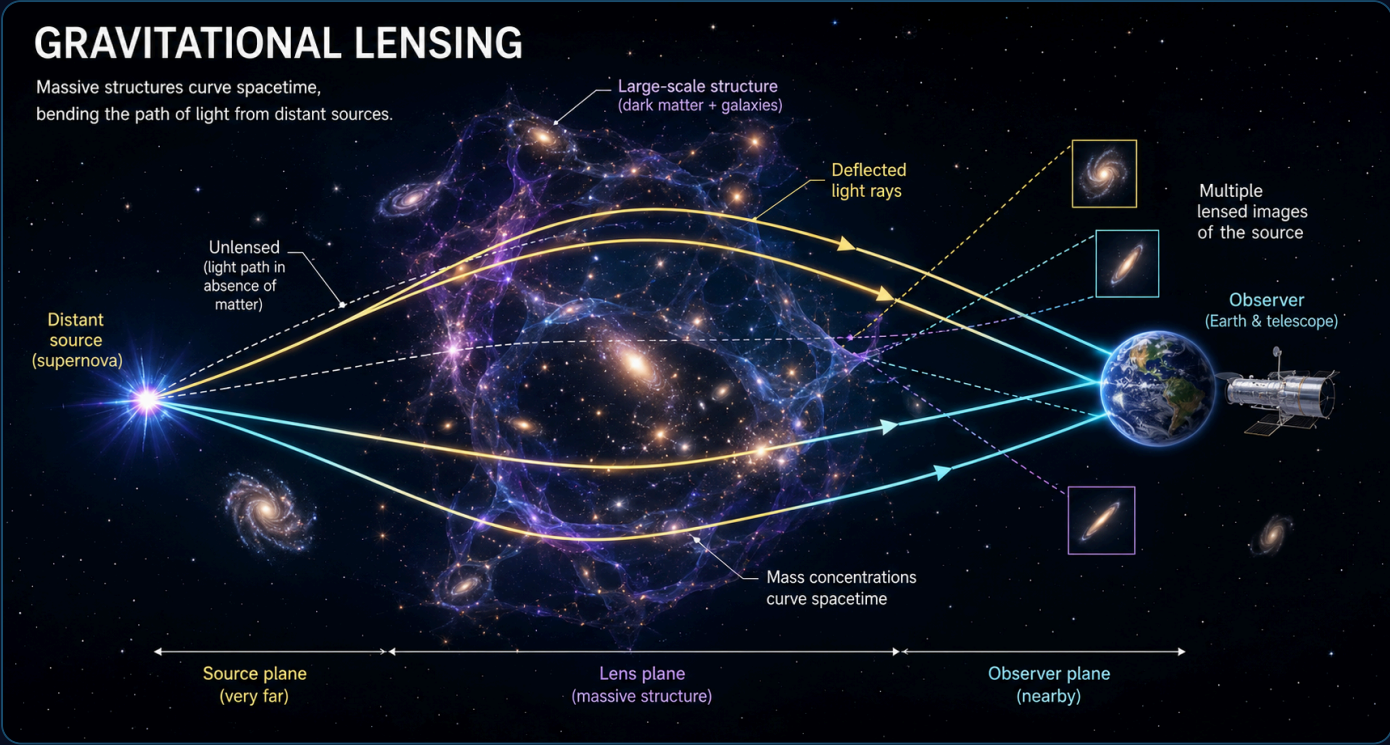
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GRAVITATIONAL LENSING



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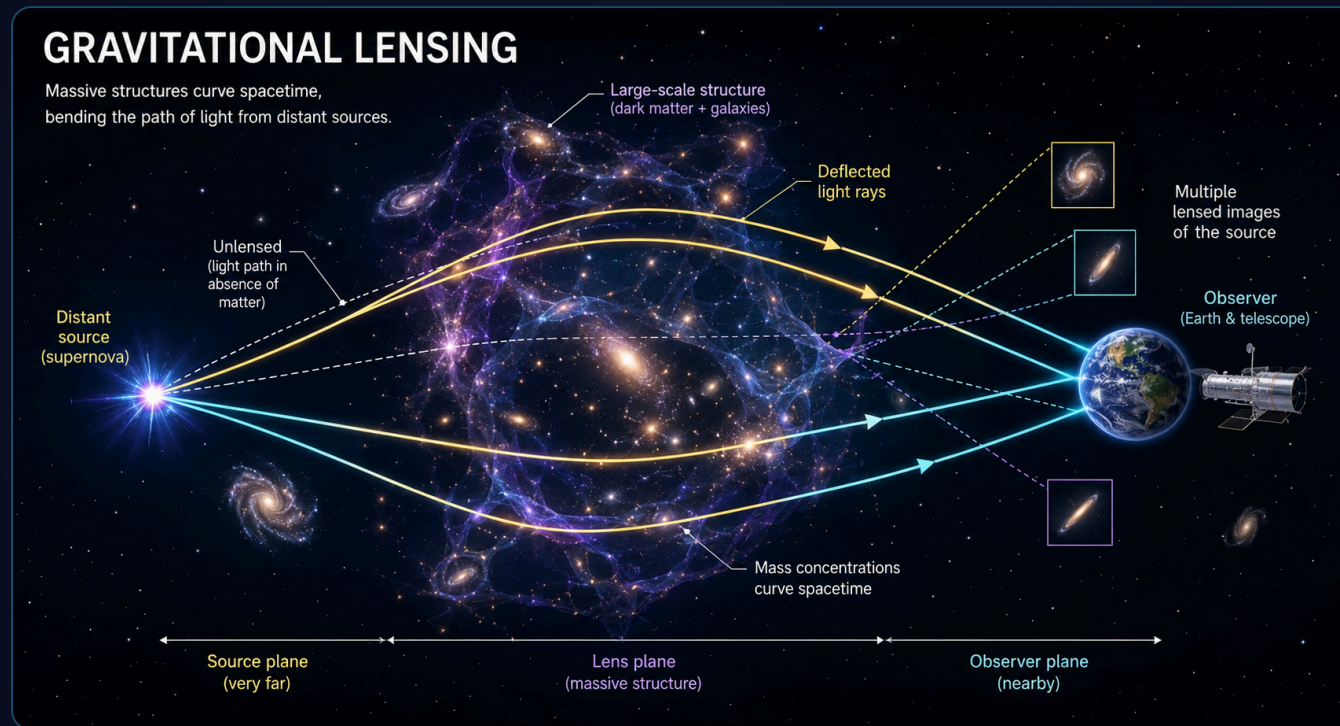
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PROBLEM

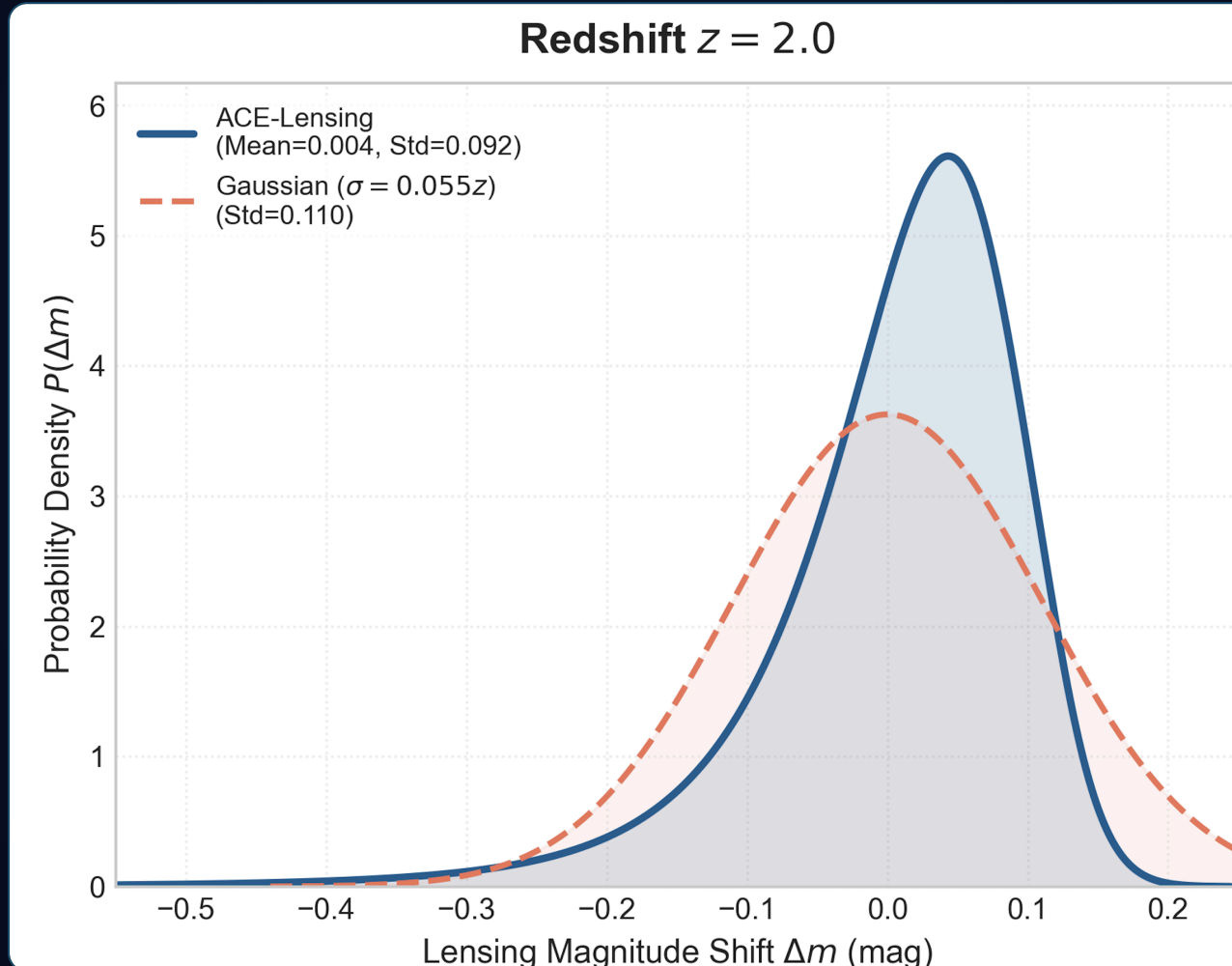
NOISE OR MEANINGFUL DATA?

Gravitational lensing **distorts** the light from distant objects.

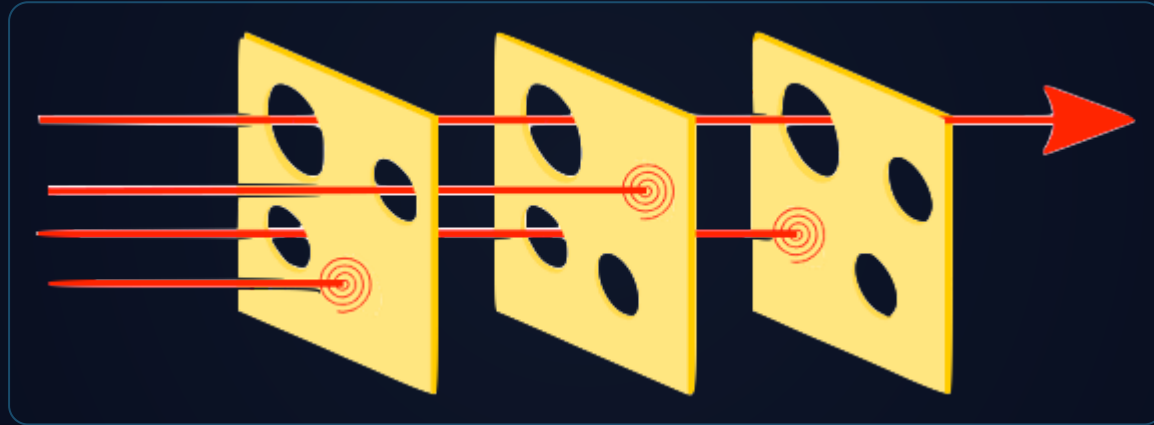


How does this lensing effect depend on cosmology?

GAUSSIAN TREATMENT

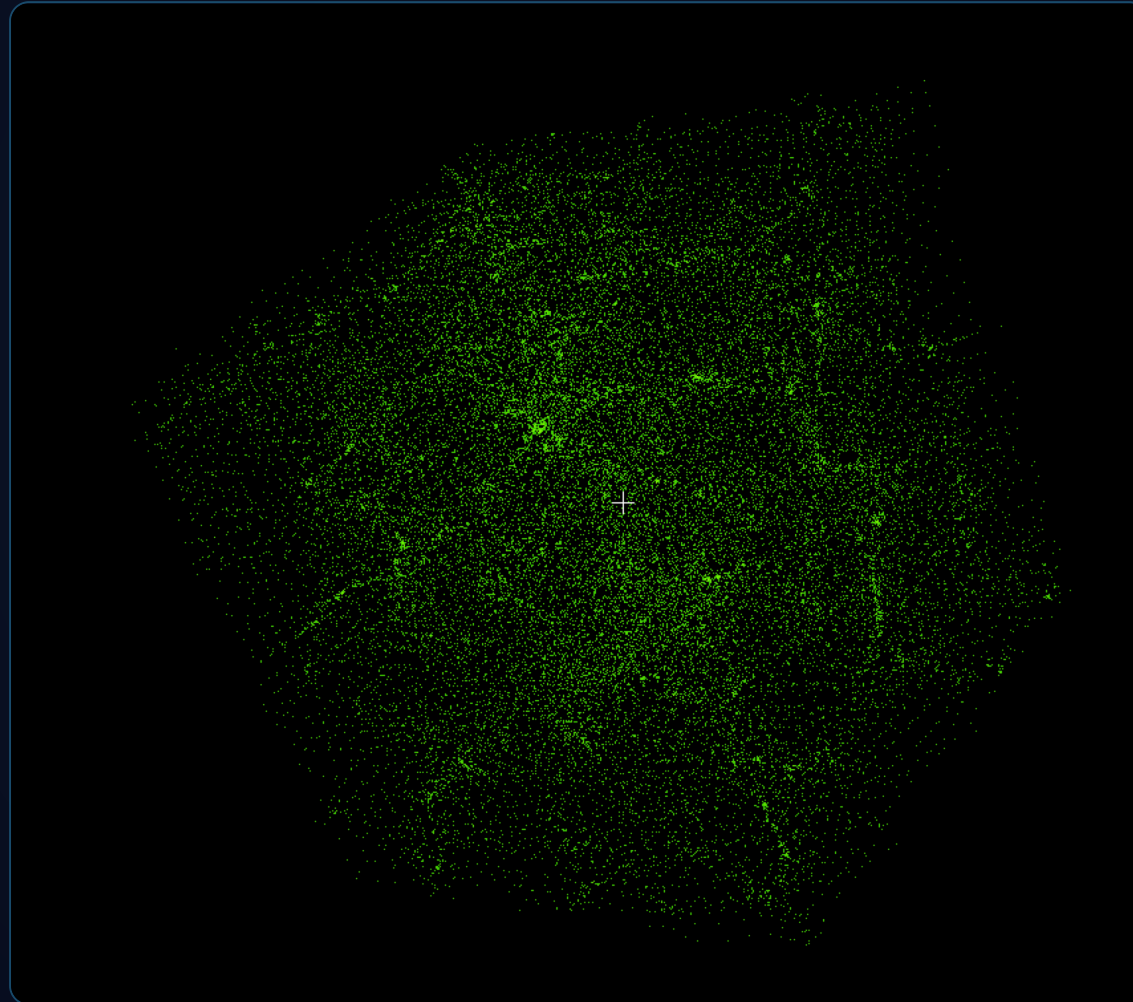


THEORETICAL MODELING

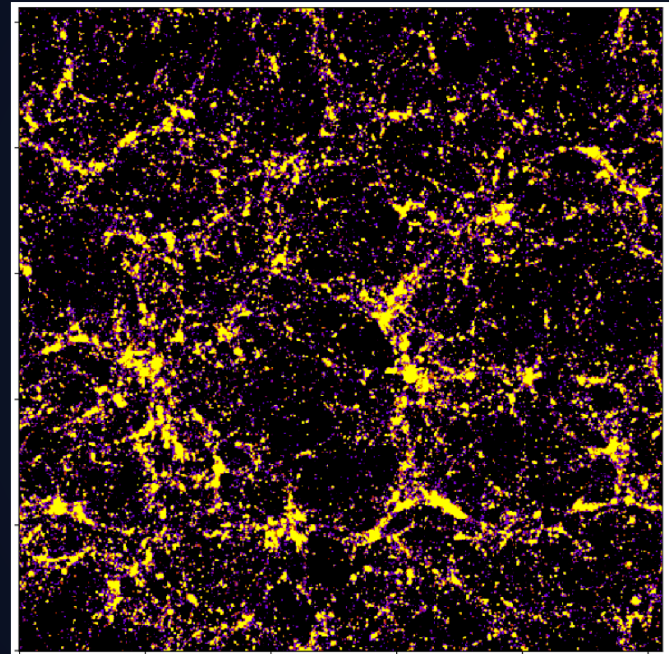


METHODOLOGY

SIMULATIONS



SIMULATIONS



METHODOLOGY

SIMULATIONS

24 hrs

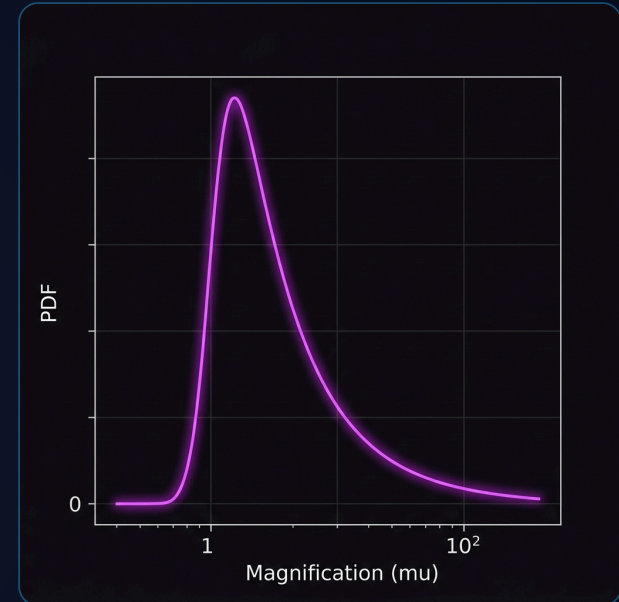
Cost of one N-body
simulation

1.7k

Supernovae in Pantheon+

1M+

Likelihood calls in MCMC



METHODOLOGY

EMULATOR

PARAMETERS

$\Omega_m \rightarrow (0.2 - 0.4)$

$h \rightarrow (0.6 - 0.8)$

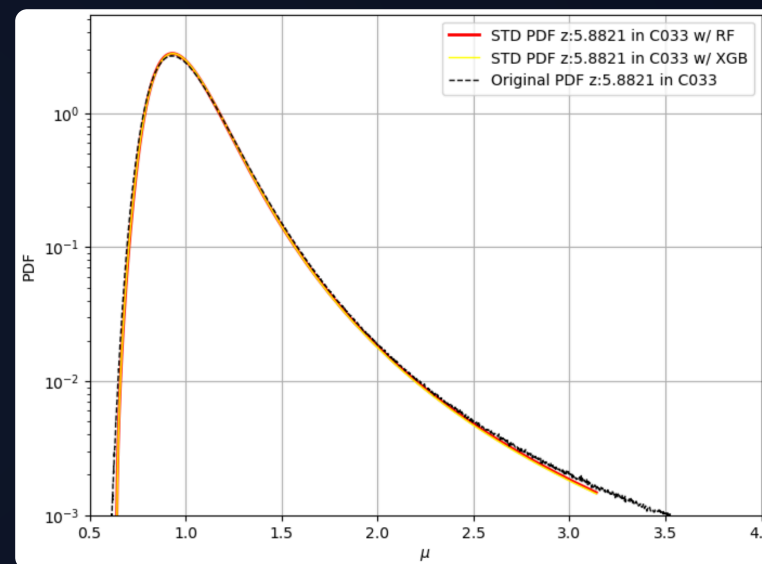
$w \rightarrow (-1.5 - -0.7)$

$\sigma_8 \rightarrow (0.7 - 0.9)$

$z \rightarrow (0.3 - 6)$



STATISTICS



70

Different cosmologies

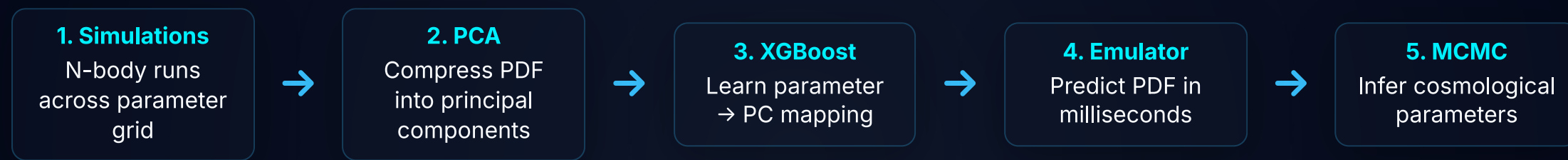
1.5k

PDFs

APPROACH & REASONING

HOW DID WE TACKLE IT?

Instead of calling the simulation directly, we **train a machine learning emulator** that learns the mapping from cosmological parameters → lensing PDF. This replaces a **24-hour simulation** with a **millisecond prediction**.



APPROACH

EMULATOR

PARAMETERS

$$\Omega_m \rightarrow (0.2 - 0.4)$$

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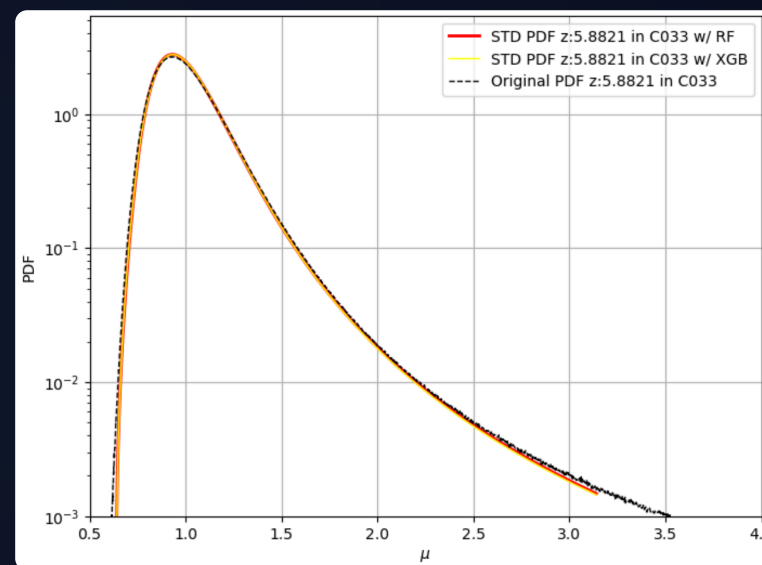
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APPROACH

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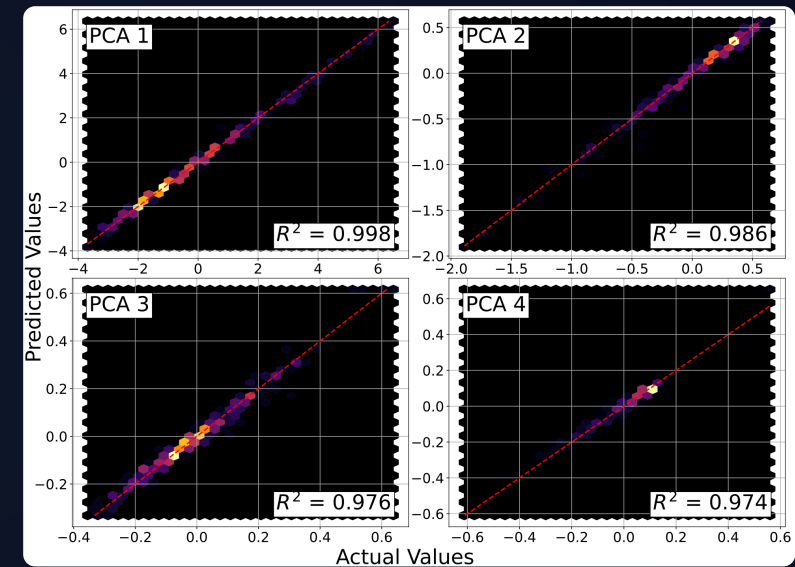
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4 PCA COMPONENTS



70

Different cosmologies

1.5k

PDFs

REASONING

TRADITIONAL METHODS VS EMULATOR-BASED METHOD

	Traditional Methods	Emulator-Based Method
Use Case	Detailed modeling	Fast parameter estimation
Computational Cost	High	Low
Speed	Slow	Fast
Flexibility	Depends on simulation or model	High
Accuracy	High	High

REASONING

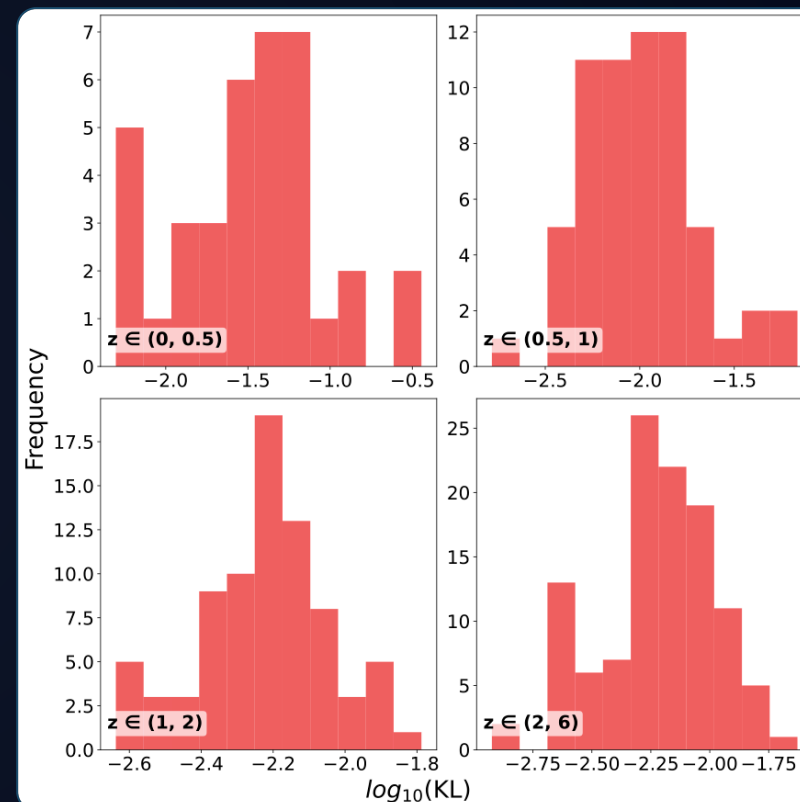
WHY XGBOOST + PCA?

WHY PCA FIRST?

- ◆ Lensing PDFs are **high-dimensional** vectors
- ◆ PCA compresses them into **4 components** capturing **99%** of variance
- ◆ Predict components independently

WHY XGBOOST?

- ◆ Handles **non-linear** parameter dependencies well
- ◆ Fast to train and predict
- ◆ Robust with **limited training data** (simulation budget)



ACE-LENSING

ACE-Lensing is a full end-to-end pipeline: from N-body simulation outputs to cosmological parameter inference.

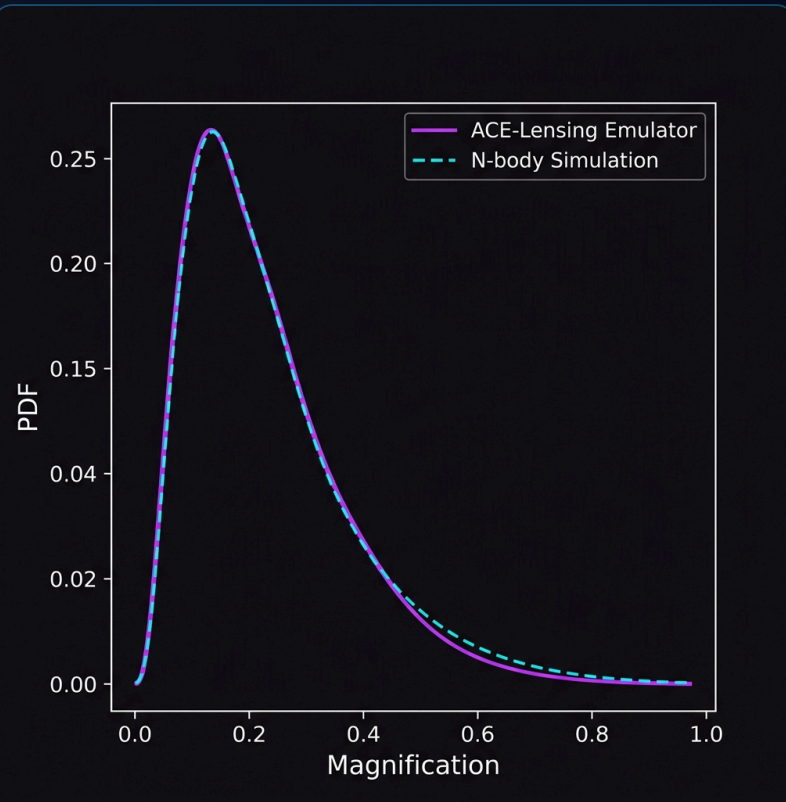
Input: Cosmological parameters (Ω_m , σ_8 , w , h)



ACE-Lensing: XGBoost + PCA emulator



Output: Magnification PDF in **~1 ms**



MCMC LIKELIHOOD MODELS

1. GAUSSIAN (FIXED)

$$\mathcal{L}_i = \mathcal{N}(\Delta\mu_i; 0, \sigma_i^2)$$

$$\sigma_i^2 = \sigma_{\text{int}}^2 + \sigma_{\text{meas},i}^2 + \sigma_{\text{lens,fid}}^2(z_i)$$

2. GAUSSIAN (ACE)

$$\mathcal{L}_i = \mathcal{N}(\Delta\mu_i; 0, \sigma_i^2(\theta))$$

$$\sigma_i^2(\theta) = \sigma_{\text{int}}^2 + \sigma_{\text{meas},i}^2 + \sigma_{\text{lens}}^2(z_i, \theta)$$

3. EDGEWORTH EXPANSION

$$p(\Delta\mu_i) \approx \mathcal{N}(\Delta\mu_i; 0, \sigma_i^2) \left[1 + \frac{\kappa_3}{6\sigma_i^3} H_3 \left(\frac{\Delta\mu_i}{\sigma_i} \right) + \frac{\kappa_4}{24\sigma_i^4} H_4 \left(\frac{\Delta\mu_i}{\sigma_i} \right) \right]$$

$$\sigma_i^2 = \sigma_{\text{int}}^2 + \sigma_{\text{meas},i}^2 + \sigma_{\text{lens,fid}}^2(z_i)$$

4. FULL CONVOLUTION

$$\mathcal{L}_i = \int \mathcal{N}(\Delta\mu_i - \delta\mu; 0, \sigma_{\text{int}}^2 + \sigma_{\text{meas},i}^2) p_{\text{ACE}}(\delta\mu | z_i, \theta) d(\delta\mu)$$

THANK YOU

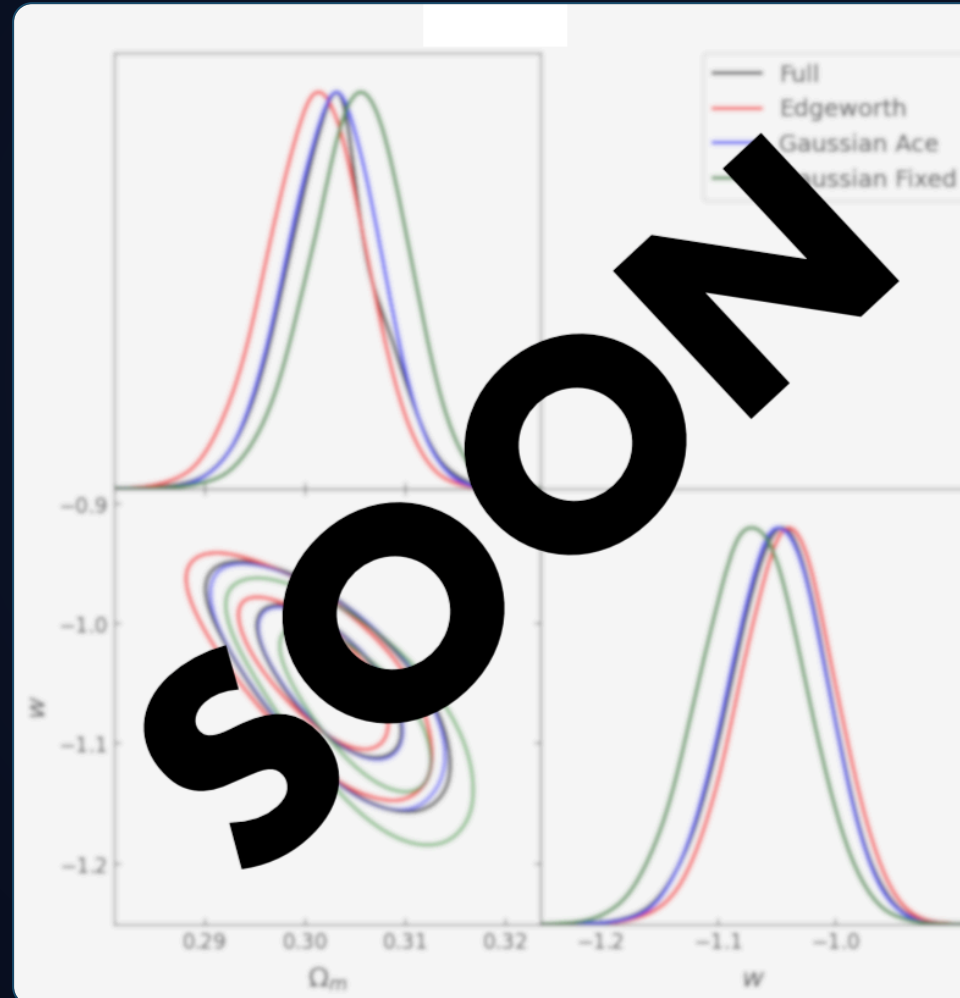
Questions?

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CONCLUSION

RESULTS



NOW

PROBLEM TO SOLVE

Decision Tree Discretization

